

AI Advice Timing and Independent Judgment in Youth

A Study Proposal Comparing Initial Judgment First, AI Advice First, and No AI Workflows

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Status *Early stage proposal. No participants have been recruited and no data have been collected.*

The question

My interest in AI began with a broader question from development economics: how do the environments people learn and work in shape their opportunities? I now examine the everyday decisions that schools and other organizations make about AI use, including when people form their own views, check advice, and receive feedback. This proposal takes one part of that question into youth learning: does the timing of AI advice change what a learner can later understand and do independently?

AI can improve the answer we see without telling us much about what the user has learned. One student may give the right answer because she checked the AI's reasoning and came to the same conclusion. Another may give the same answer simply because she accepted what the AI said. The observed result is identical, but the reasoning behind it is not.

A useful evaluation should therefore ask two separate questions. Did AI improve the work produced while the tool was available? Did it also help the student build knowledge and judgment that remain after the tool is gone? I use independent judgment in a narrow, testable sense: performance on a related task without AI, a sound response when AI advice is wrong, and confidence that reflects actual accuracy. Recording an answer before seeing AI is an intervention, not proof that independent thought occurred.

Learning research has long distinguished performance during practice from learning that persists and transfers after practice ends (Soderstrom and Bjork, 2015). Recent experiments point to the same measurement gap without settling the design question. In a randomized field experiment with nearly 1,000 high school students, a GPT-4 interface designed to resemble ChatGPT improved performance on practice problems, but students scored 17 percent lower than the no-AI group on the unassisted exam that followed. A tutor with teacher-designed safeguards largely mitigated that loss but did not produce a positive exam effect (Bastani et al., 2025).

A preregistered study enrolled 405 students aged 14 and 15 and retained 344 after applying its exclusion criteria. Three days later, both note-taking alone and note-taking alongside an LLM improved literal retention and comprehension relative to LLM use alone. Note-taking alone also improved free recall. Students generally preferred the LLM to note-taking alone and perceived it as more helpful (Kreijkes et al., 2026). In a randomized trial with 1,053 first-year undergraduates, ChatGPT raised the score on a standard marketing rubric, while causal-reasoning training increased mechanism-based and falsifiable reasoning and the diversity of ideas but did not raise that rubric score (Asirvatham et al., 2026). Better work during AI use, learning that lasts, and the user's experience of the tool can move in different directions.

Evidence on ordering is mixed, and most of it comes from adults. In a study of 19 veterinary radiologists, participants who saw AI from the start agreed with it more often, whether the AI was correct or not. On findings with the most reliable ground truth, the one-step group performed better than the provisional-judgment group when AI was correct and worse when AI was wrong. Participants who recorded a provisional judgment first changed only 5 percent of the findings on which their initial judgment disagreed with the AI (Fogliato et al., 2022).

In separate nutrition experiments, simulated, always-correct AI recommendations improved immediate decisions when shown before a choice and when shown after an initial choice. Neither design, however, produced greater learning than the minimal-feedback baseline on subsequent questions in the same session. An explanation without an explicit recommendation did produce a learning advantage. Because the two orders were tested in separate experiments, these results do not show that either sequence is generally better (Gajos and Mamykina, 2022). The information an interface provides may matter as much as when it appears.

A June 2026 essay by Tamara Tate and Mark Warschauer proposes a Human-First AI framework. Its "Think First" habit asks learners to form a brief idea or plan before consulting AI, while later habits emphasize checking outputs and reflecting on the learning process. The framework offers a practical

rationale, not causal evidence that this sequence improves learning. The proposed study would test one part of that rationale by holding AI advice constant and measuring later unaided performance and responses to misleading advice (Tate and Warschauer, 2026).

That leaves two live possibilities. Seeing AI advice first may support a novice by supplying useful information, but it may also anchor the learner to a plausible mistake. Recording an initial judgment first may create a reference point for evaluating the advice, but it may also anchor the learner to an early error. These are competing hypotheses, not findings about young people. The studies reviewed here do not establish which sequence better supports later unaided performance. The proposed study is designed to test that question rather than assume the answer.

These questions fit a broader emerging effort to measure whether AI strengthens learning and durable human capability across settings. OpenAI's Learning Lab is bringing researchers, educators, students, and builders together to develop shared questions and measures across contexts. This proposal takes on one narrower question: whether the timing of AI advice changes what learners can later do without AI and how they respond when AI is wrong (Devani and Leiva, 2026).

A practical test

I would begin with a small feasibility study run through an institution experienced in research with adolescents. Everyone would first complete the same short task without AI. Participants would then be randomly assigned to one of three ways of working. The age range, task, setting, and sample size should follow pretesting and consultation with research partners.

Workflow	What the participant does
Initial Judgment First	Sees the task, records an initial answer, confidence, and brief reason, then sees prepared AI advice and may revise.
AI Advice First	Sees the task with the same prepared AI advice before recording an answer, confidence, and brief reason.
No AI	Sees the task without AI, records an answer, confidence, and brief reason, and may review it independently.

All three groups would receive the same cases and verified feedback after submitting their final answers. Feedback content and timing would follow the same rules. The two assisted groups would also receive the same prepared advice through an interface that differed only where the workflow required it. Some advice would be correct and some plausible but wrong, with versions rotated across participants. Participants would know that the AI may be wrong. Prepared advice would support a controlled comparison, although it would not reproduce a full chatbot conversation.

About one week later, everyone would first complete new, related tasks without AI. Only after this unaided test would the assisted groups receive additional tasks with correct and misleading advice. This order protects the main measure of what participants can do on their own.

Primary outcome Accuracy on the later tasks completed without AI.

Secondary outcomes Immediate accuracy, final decisions under correct and misleading advice, confidence relative to accuracy, explanation quality, time, perceived effort, and acceptability.

Analysis Register the main outcome and comparisons in advance, account for initial performance, use graders who do not know the assigned workflow, and report uncertainty rather than only whether a result crosses a significance threshold.

Because the participants would be minors, this should not be a solo study. It would require an institutional partner, ethics review, parental consent, youth assent, low-risk tasks, minimal data collection, corrective feedback, and a clear debrief. No identifying information should be sent to a model. Educators, young people, child development researchers, statisticians, and research ethics staff should review the tasks and procedures before recruitment.

What the evidence could and could not show

The study would measure current performance, later unaided performance, and decisions under misleading advice separately. A promising result for either workflow could guide a larger test, but a small pilot would mainly establish whether the tasks and procedures work. Similar estimates would remain inconclusive if uncertainty were too large to rule out meaningful differences. Even a clear difference would not explain its cause on its own: for example, better performance with advice first would not establish that an initial judgment anchors people to a mistake.

The study would test a whole workflow, not isolate a single psychological mechanism. Initial Judgment First changes timing, effort, and commitment together. Prepared, single-turn advice is not a full chatbot experience. A result after one week would not establish long-term development, and one task, age group, or setting could not represent all youth AI use. The narrower aim is to learn whether a larger study is warranted and which outcomes it should measure.

For a school, the practical question is when AI advice should appear and what students should still have an opportunity to work out themselves. Related questions arise in workplaces, where people need to build expertise while completing tasks. This youth study would not establish effects on adult professionals or on an organization's ability to learn from decisions. Those would require separate studies. My contribution here would be evidence review, task and survey design, implementation planning, data checks, analysis, and documentation, working with partners who understand young people and their learning settings.

Selected references

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